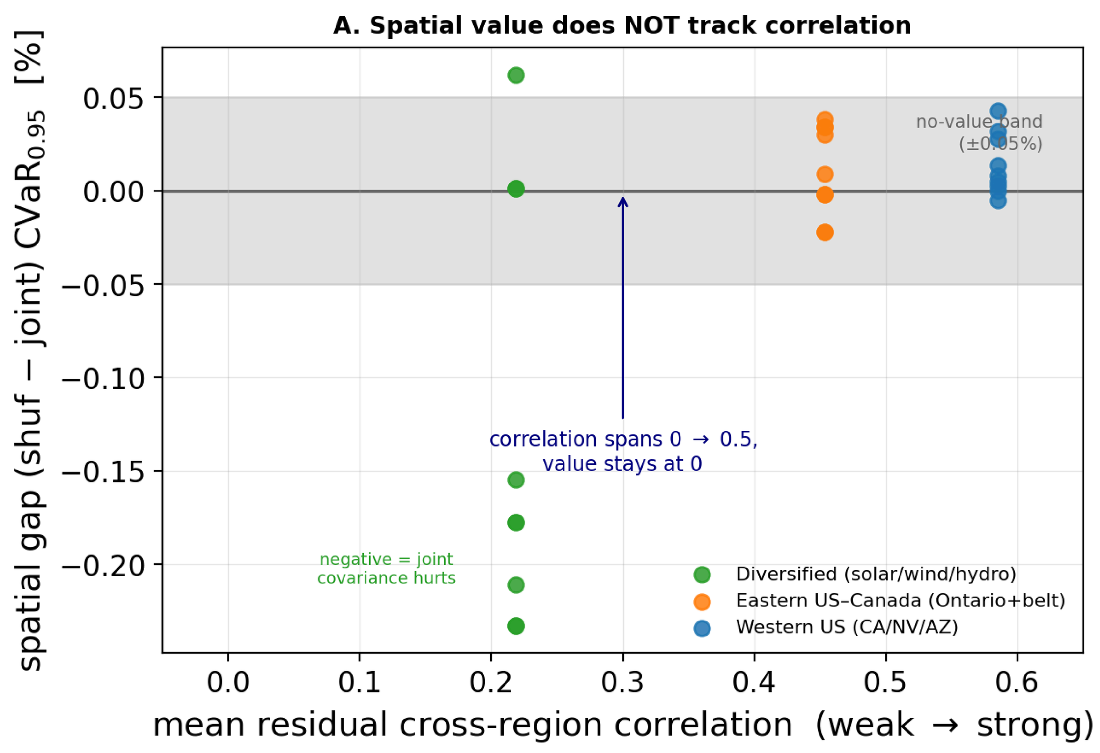
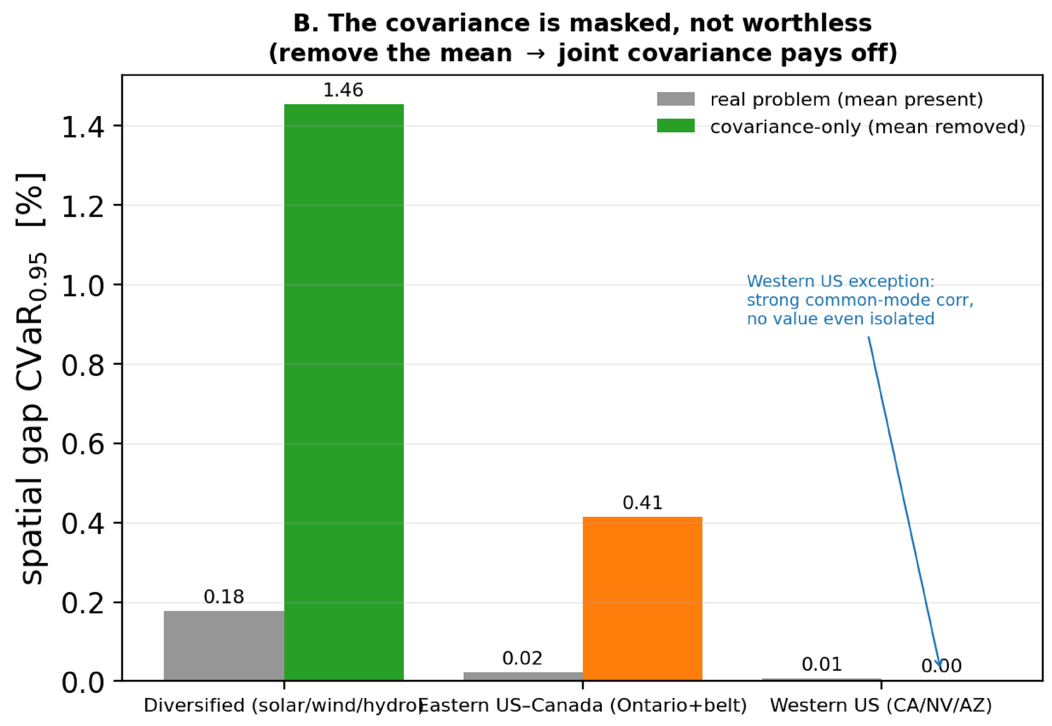
Does Spatial Correlation of Carbon Intensity Improve Robust Carbon-Aware Data-Center Scheduling?

A multi-grid falsification study: a causal mechanism, a copula test, and an active-transfer extension

MARCO ORTIZ TOGASHI · SUPERVISOR: PROF. BISSAN GHADDAR · IE UNIVERSITY, MASTER IN BUSINESS ANALYTICS & DATA SCIENCE





THE FINDING

Spatial correlation of carbon is *real*, but a robust scheduler can't use it.

Across every grid and every dependence model, the out-of-sample spatial gap never leaves **zero**. The value lives in the diurnal **mean** carbon field, not in how regions correlate.

0.78	~0%	+1.46%	4–10%
CROSS-GRID CORRELATION IS REAL	ROBUST SCHEDULING VALUE	ONLY WHEN THE MEAN IS FLATTENED	ONLY VIA ACTIVE TRANSFER

The question

Compute is unusually **flexible**: batch and training jobs can shift in time and **across sites** (logically, over the network, no grid interconnection needed). Carbon intensity varies by hour and region, so a scheduler can defer work to cleaner moments under **distributionally robust optimization (DRO)**.

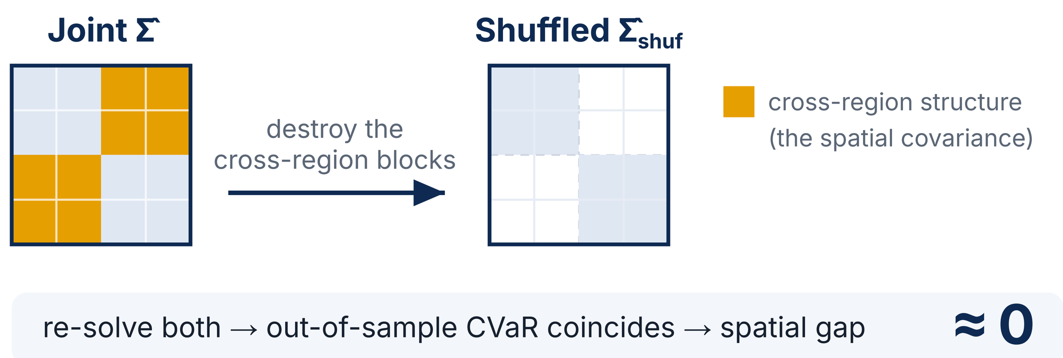
Prior models treat each region in isolation. We treat carbon intensity as a stochastic **vector** so that **spatial correlation** can inform the schedule, and ask whether it helps.

How we test it

A Mahalanobis–Wasserstein DRO, solved as a second-order cone program:

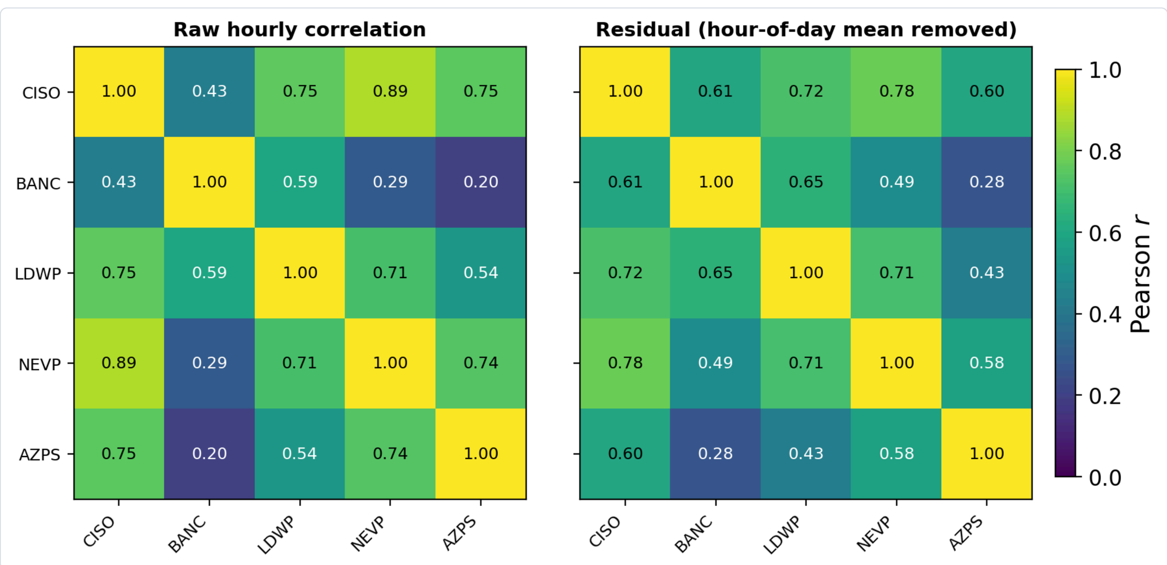
$$\min_{x\in\mathcal{X}} \langle \bar{\rho}, x\rangle \;+\; \varepsilon\,\|L^\top x\|_2, \qquad LL^\top = \widehat{\Sigma}$$

The shuffled-marginals test. Zero the cross-region covariance blocks, re-solve, and compare out-of-sample $\mathrm{CVaR}_{0.95}$. A positive gap would mean correlation helps.



- **Pre-registered**: config commit-locked, single 2025 test read.
- Three US/Canada grids spanning low-to-strong correlation ($\bar{r} \approx 0.2$ to 0.65).
- Best-case stress test: real fleets are dispersed (low correlation) and also null.

The premise holds



Residual cross-region correlation is strong and survives de-seasonalization (US West up to 0.78). The spatial assumption is empirically valid.

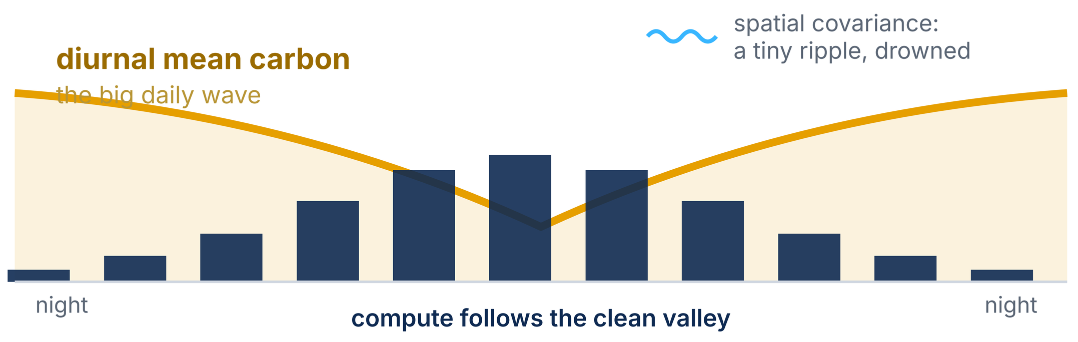
It adds no robust value

Across the spectrum the spatial gap **never exceeds a few tenths of a percent** and is not sign-stable. It survives shrinkage, residualization, Benjamini–Hochberg over 144 cells, walk-forward, tightness sweeps, and the full $\mathrm{CVaR}_{0.90}$–$\mathrm{CVaR}_{0.99}$ tail range.

Statistically zero, not undetected. A per-cell equivalence test (TOST) rejects any effect above the materiality margin in all 27 cells. And the DRO is genuinely *active* ($\varepsilon^\star = 1$ almost everywhere), it pays for robustness and the covariance still buys nothing.

Why, the mechanism

Mean-ablation (causal). Flatten the mean field and the joint covariance suddenly pays, up to **+1.46%**: the spatial signal is *real but masked* by the diurnal mean, not absent.



The schedule chases the big mean valley; the covariance ripple is too small to move it.

Tail dependence (structural). The residual dependence is also *non-elliptical*: regions go clean together more than dirty together ($\chi_L > \chi_U$), so an elliptical covariance ball is the *wrong object* as well as a subordinate one.

A mean-dominance bound

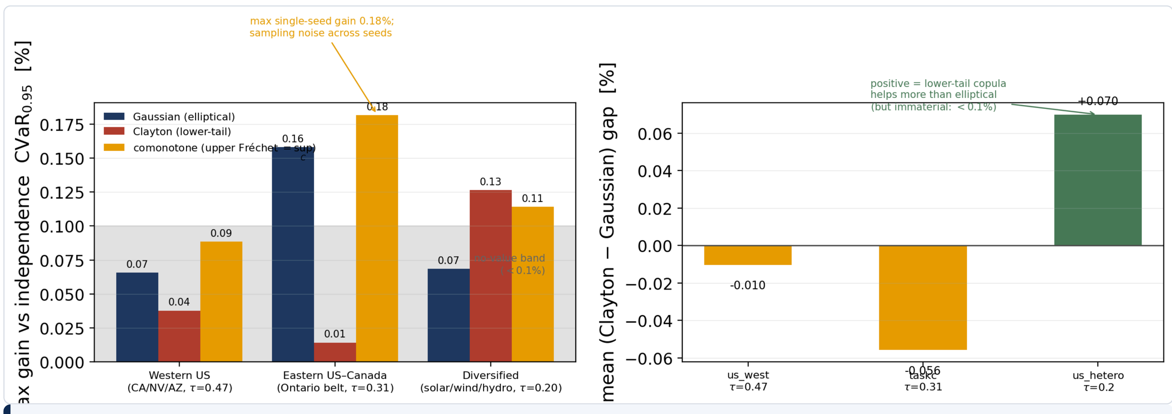
By translation invariance of CVaR the dependence model moves only the residual term, and the spatial gap is bounded a-priori:

$$\left|\mathrm{OPT}(\Sigma) - \mathrm{OPT}(\Sigma^{\mathrm{shuf}})\right| \;\le\; \varepsilon\left(\max_x \|x\|\right)\|\Sigma^{\mathrm{off}}\|^{1/2}$$

It vanishes with no robustification ($\varepsilon \to 0$) or no cross-structure. The tight ceiling is *measured* by the comonotone arm, an order of magnitude below the mean-driven value.

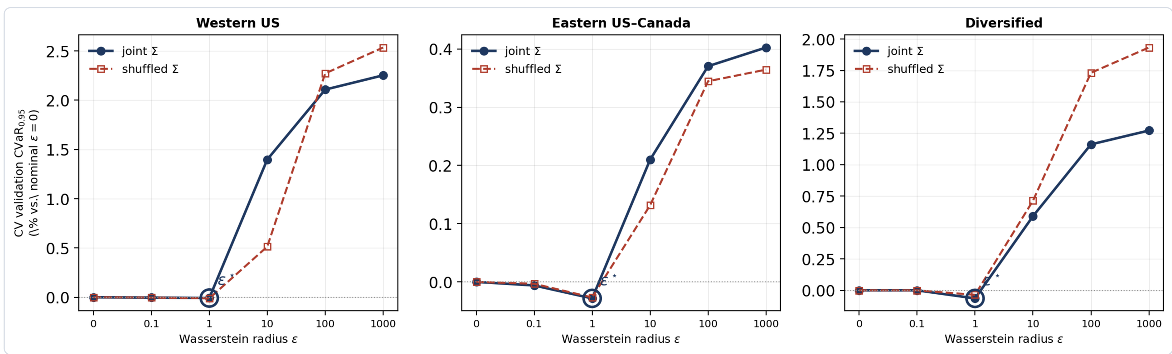
Even the right copula fails

We replace the covariance ball with copula schedulers that *can* see the asymmetry: independence, Gaussian, lower-tail **Clayton**, and the maximal **comonotone** ceiling.



Even the *maximal* copula sits at the scenario-noise floor ($|\Delta| \lesssim$**0.1%**): the null is **total**.

The robustness is genuinely on



Blocked CV selects $\varepsilon^\star = 1$ in nearly every cell: the DRO is actively robustifying, so this is a true *no-value* result, not an under-tuned model.

Conclusions & extension

- Spatial correlation is **real but adds no robust value**, for any passive dependence model.
- The binding constraint is the **mean field** (bound + comonotone ceiling).
- **Use a per-region marginal scheduler**: it captures essentially all the value.
- An **active transfer channel** recovers **4–10%** *deterministically*, but robustifying it pays nowhere data-grounded (online or under real emergencies).

Reproducible: 187 unit tests, pre-registered configs, equivalence tests, multi-seed stability, an independent review. AI used as a coding/drafting assistant under the author's direction.

Seen directly: the joint and shuffled-covariance schedules coincide (US West)